DevOps Architecture

Layers, Integration & Pipeline View

Presented By:

Vaani Jindal - 500119144
Harshit Kumar - 500122407
Arnav Sharma - 500120678

Course Name : Containerization and Devops

Faculty Name: Dr. Prateek Raj Guatam

Introduction to DevOps

DevOps refers to the merger of Development and Operations.

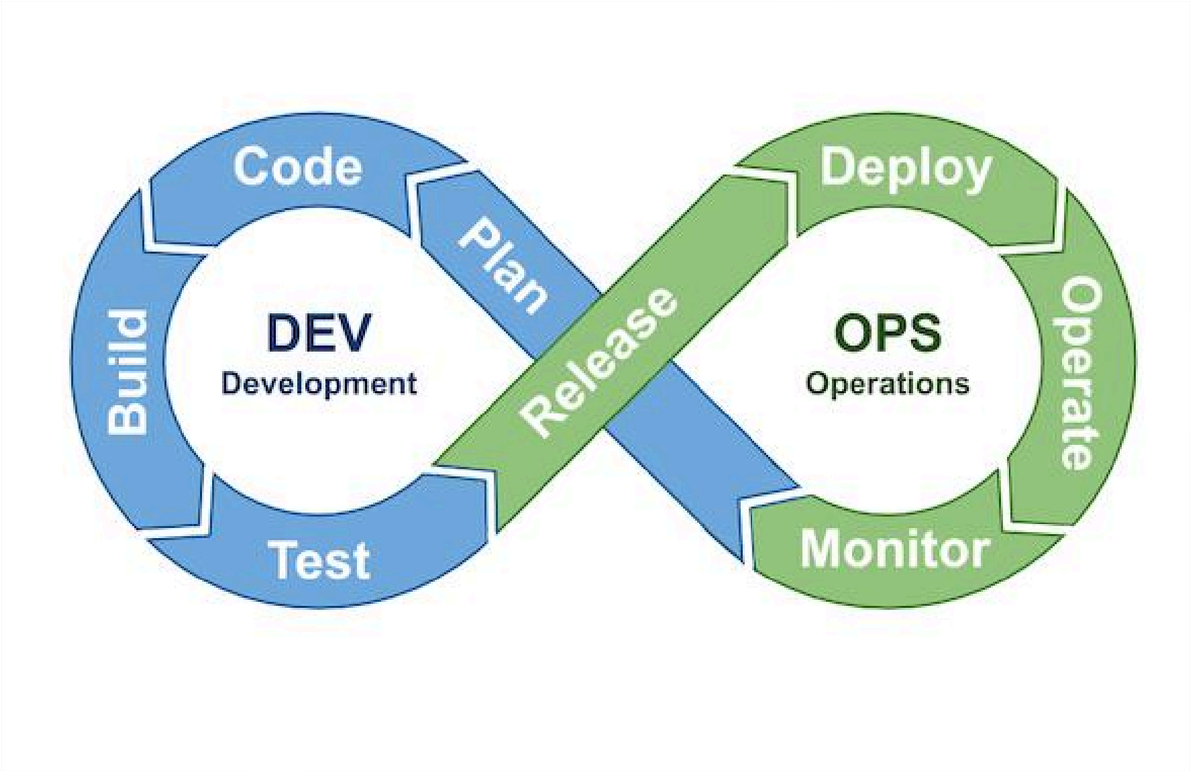
DevOps is a contemporary methodology in software development that ensures:

- Faster software development
- Efficient deployment of changes
- Less number of errors
- Better collaboration

Previously, developers and operations worked independently.
The result was:

- Slow deployment
- Miscommunication
- Regular failure

DevOps addresses this challenge by working together.



What is DevOps Architecture?

1

The DevOps Architecture refers to the framework for:

- Development of software
- Testing of software
- Deployment of software
- Monitoring of software

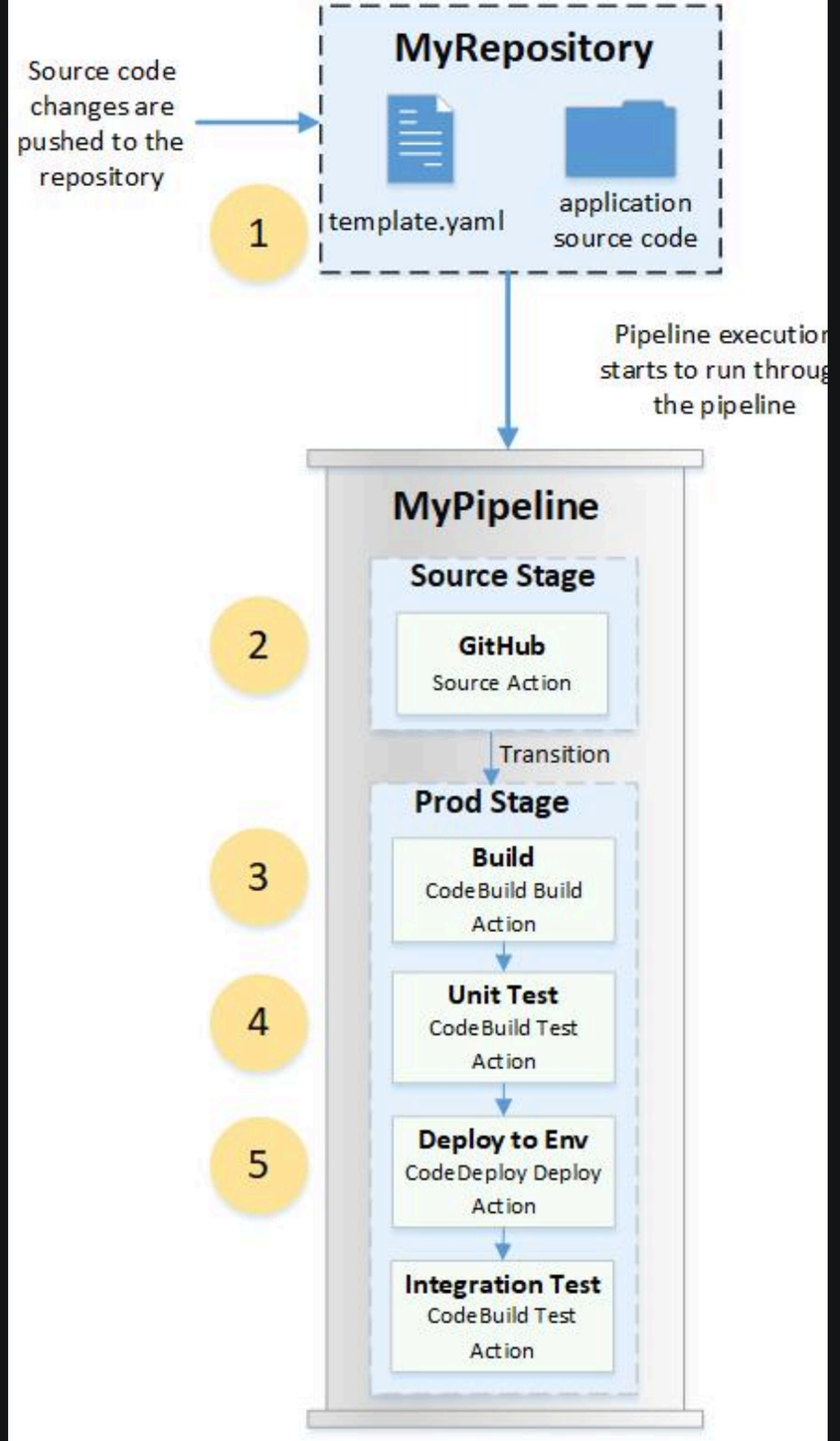
This is the process by which various tools, teams, and processes come together as one.

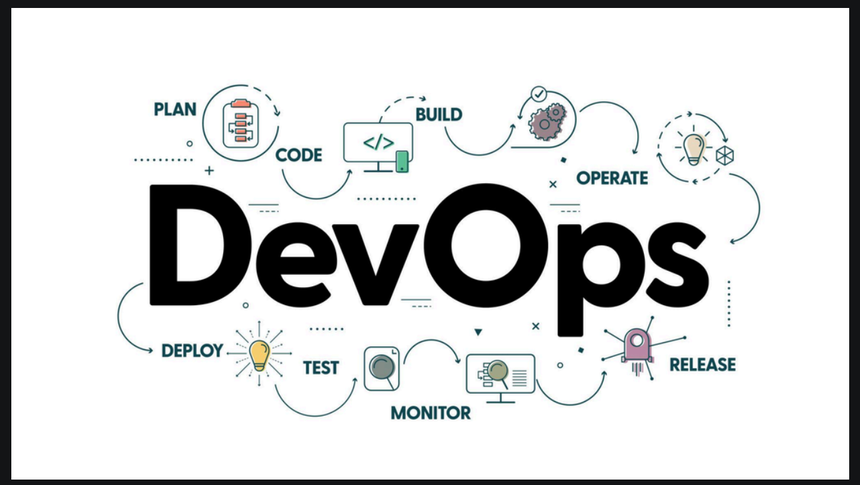
2

Purpose of DevOps Architecture

- Fast deployment
- Constant update
- Automation
- Collaboration
- Highly reliable

Why DevOps Architecture is Important?





Problems Associated with Traditional Software Development:

- Long software development time
- Manual testing
- Manual deployment
- High bug rate
- Ineffective communication

Advantages of DevOps Architecture:

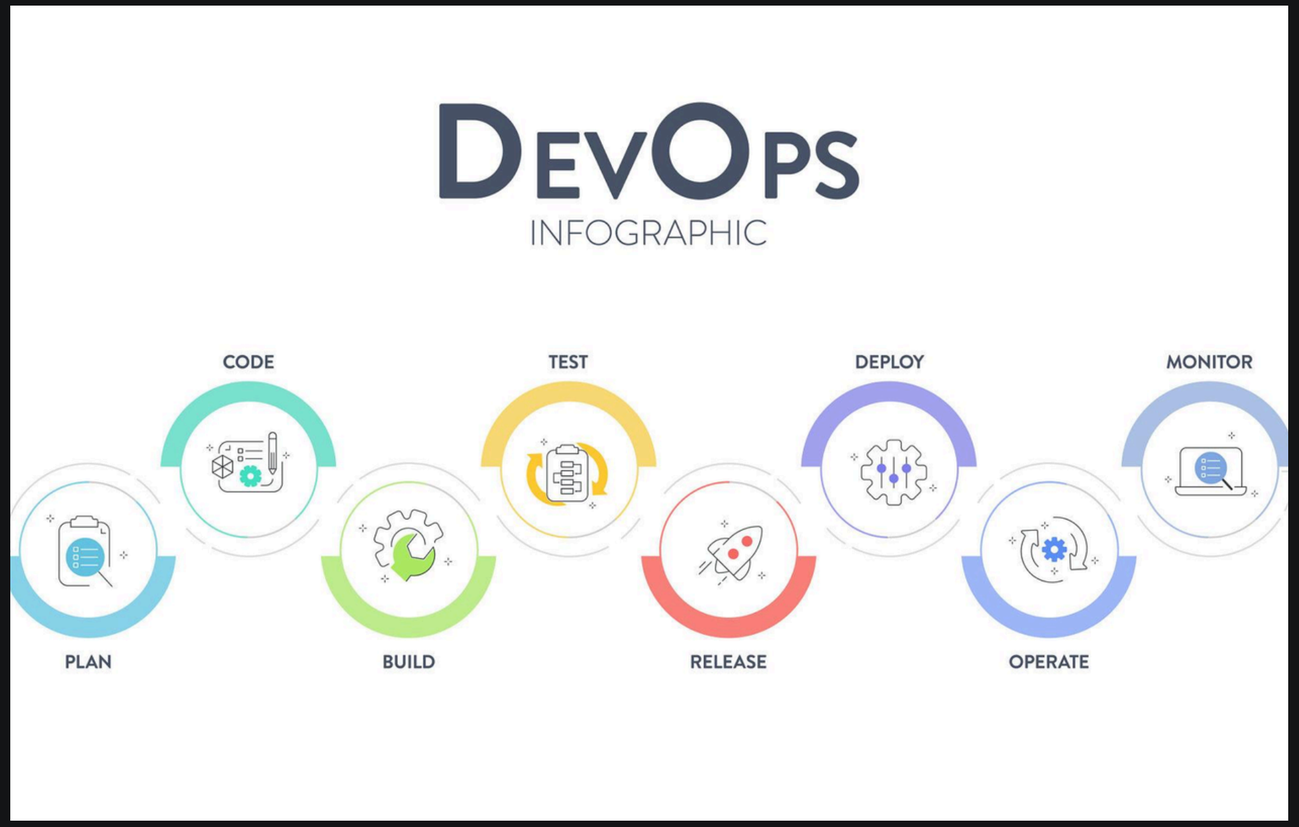
- Automation in testing
- Continuous deployment
- Effective bug resolution
- Enhanced teamwork
- Increased customer satisfaction

DevOps Architecture Overview

DevOps Architecture involves a Continuous Process Cycle:

- Planning
- Development
- Building
- Testing
- Releasing
- Deployment
- Monitoring
- Feedback

The above process is repeated continually.



DevOps Architecture Layers

DevOps Architecture consists of multiple layers:

- Application Layer
- Infrastructure Layer
- CI/CD Layer
- Monitoring Layer
- Security Layer
- Collaboration layer

Application Layer

This layer includes:

- Application Code
- Microservices
- APIs
- Containers

Code is written by developers in this layer.
Tools Used:

- GitHub
- GitLab
- Docker
- Kubernetes

Purpose:
To develop and manage applications effectively & efficiently.

Infrastructure Layer

This layer includes:

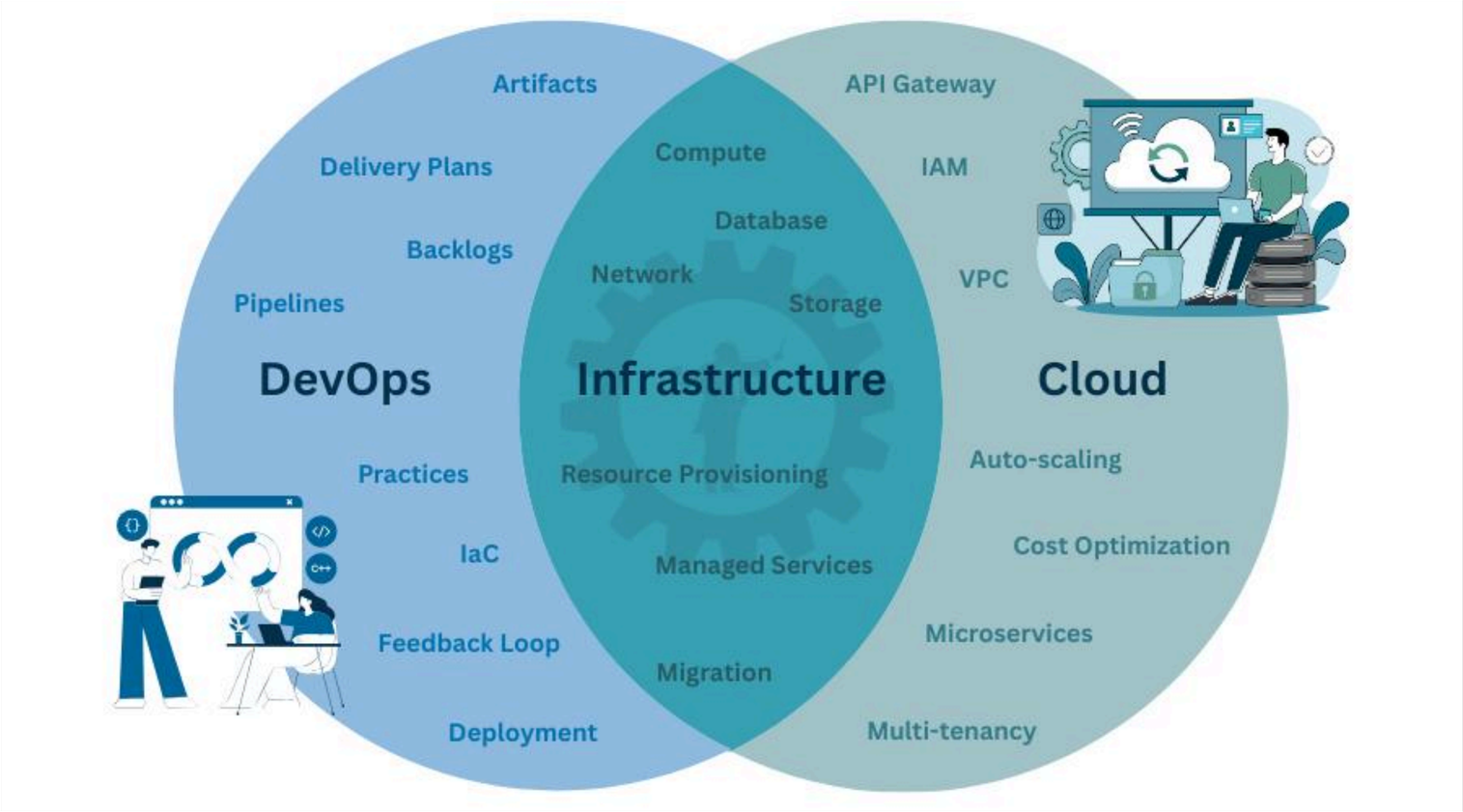
- Servers
- VMs
- Containers
- Cloud Platforms

Common Tools:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform
- Terraform

Purpose:

For scalable and flexible infrastructure.



CI/CD Layer :

CI stands for:
Continuous Integration

Code is merged on regular basis in the main repository.
And automated builds are peofromed.

CD stands for:
Continuous Deployment

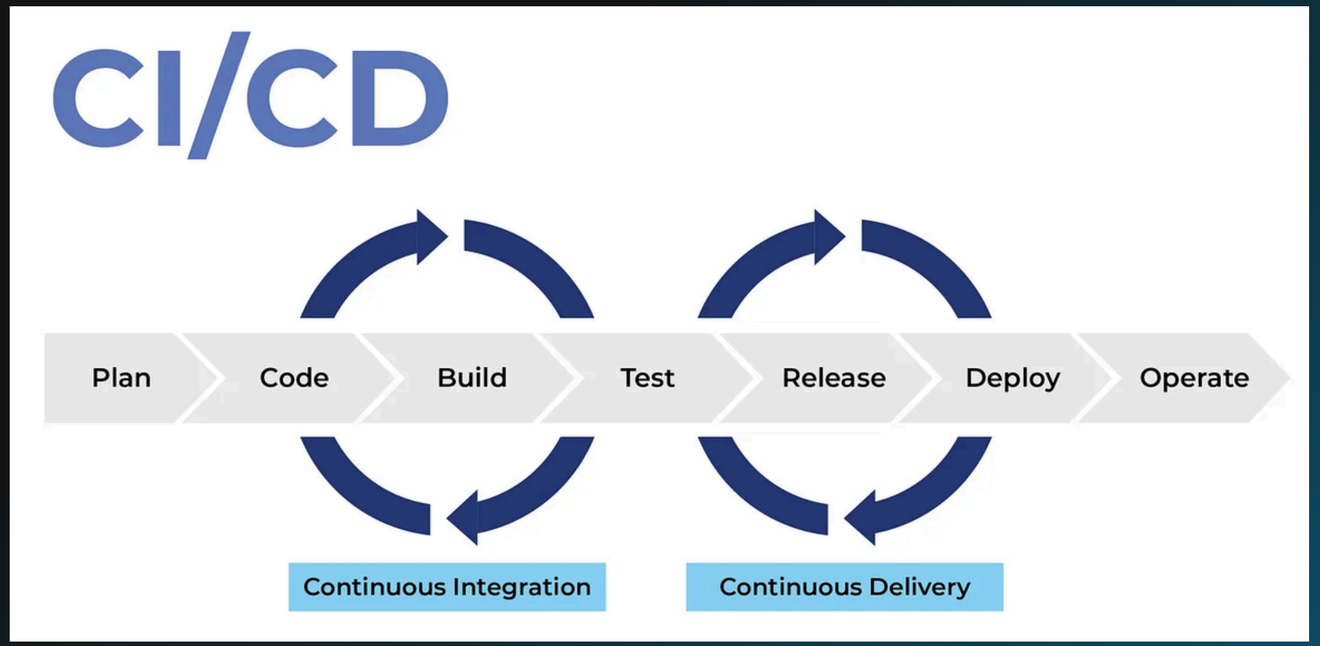
Code automatically deployed to production

Tools Used:

- Jenkins
- GitHub Actions
- GitLab CI/CD

Advantages:

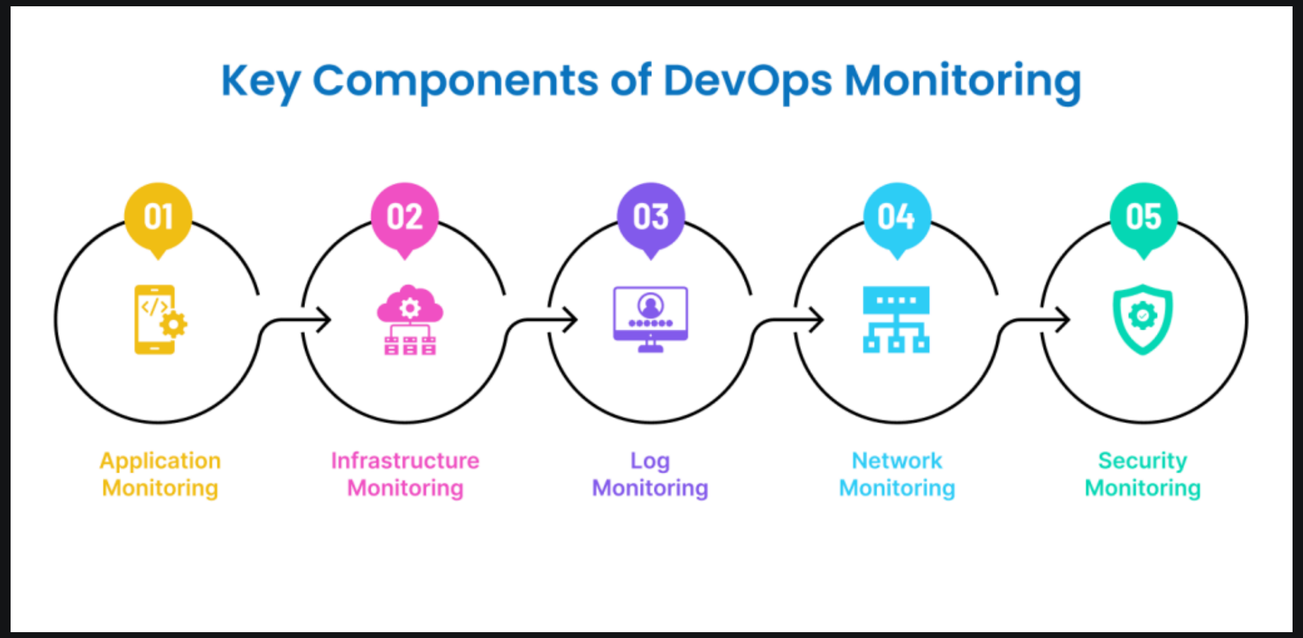
- Efficient deployment
- Reduced human errors



Monitoring Layer :

This layer monitors:

- Performance of applications
- Error/failure reporting
- Server utilization
- Activity of users



Tools Used: Aws Cloud Watch

Purpose:

Early identification of issues.

Security Layer :

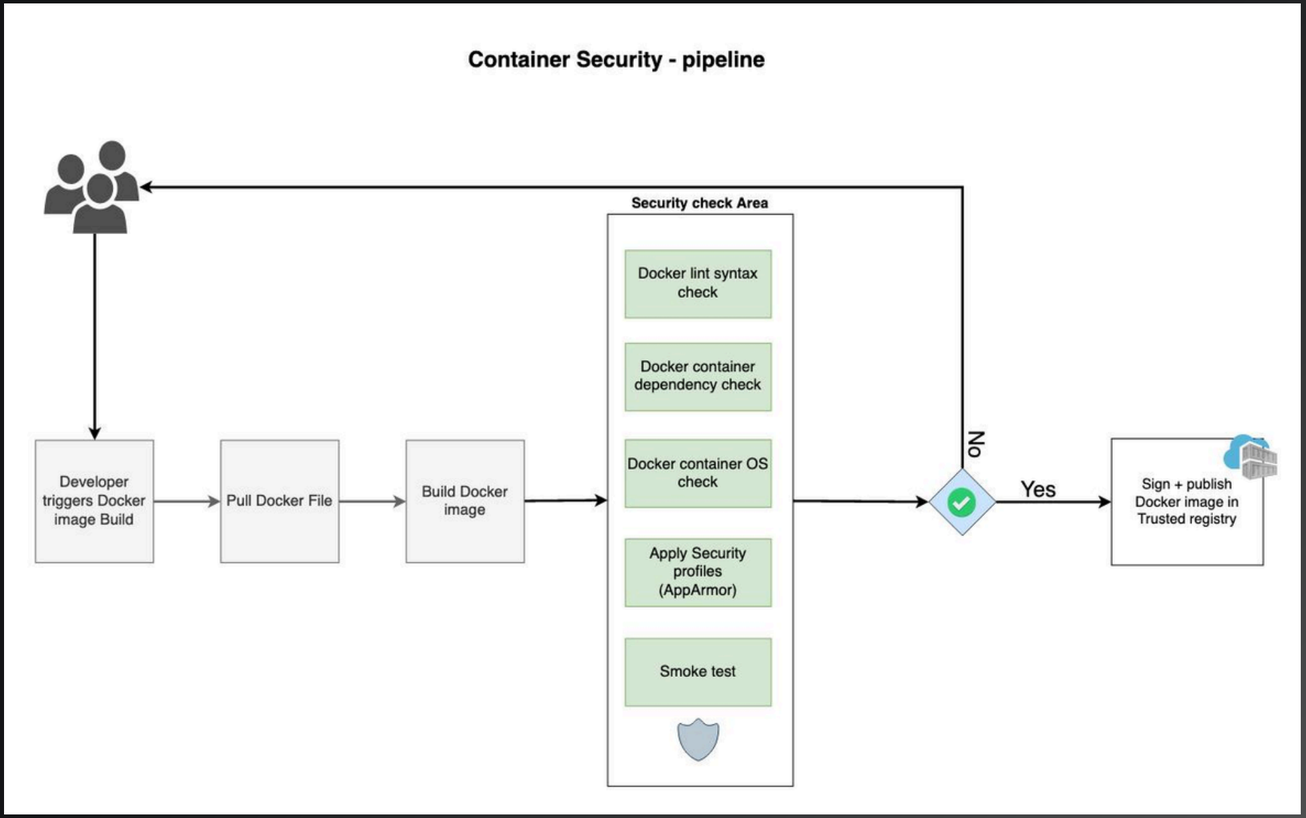
Security is included in DevOps. It is referred as DevSecOps

Security comprises:

- Code Analysis
- Vulnerability Detection
- Security Testing

Purpose:

Secured software deployment



Integration in DevOps

In DevOps there is integration of:

- Development Team
- Testing Team
- Operations Team
- Security Team

Example of tool integration is:

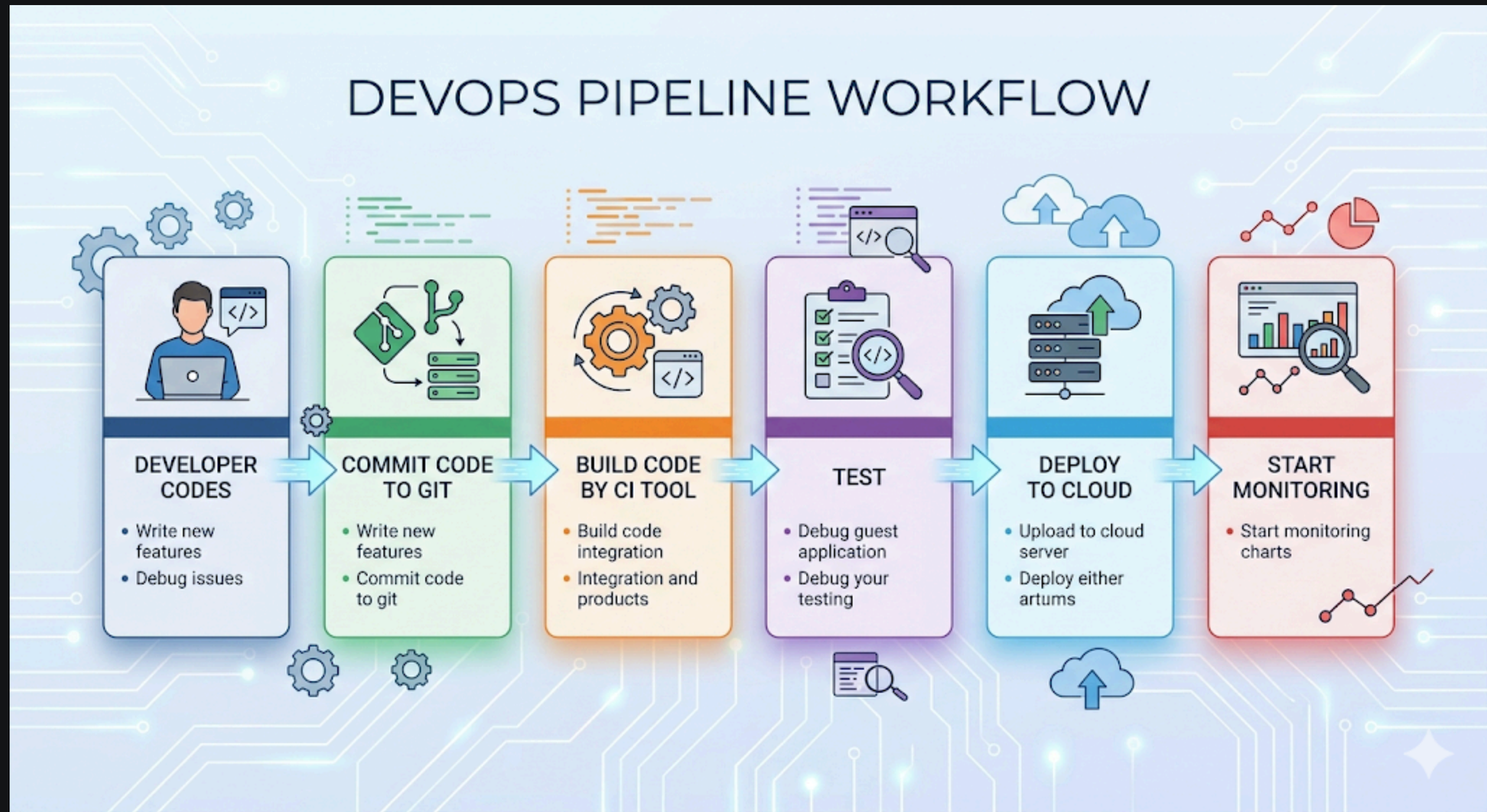
GitHub → Jenkins → Docker → Kubernetes → AWS → Monitoring

This results in automated workflow.

DevOps Pipeline View

DevOps Pipeline consists of:

- Code
- Build
- Test
- Release
- Deploy
- Monitor



Real World Example

Real-life example: Netflix

Netflix follows DevOps Architecture:

- AWS cloud
- Jenkins
- Docker
- Kubernetes

Outcome:

- Quick updates
- High availability
- No downtime

Flow:

Developer codes

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Code commit to repository

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Auto test

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Cloud Deployment

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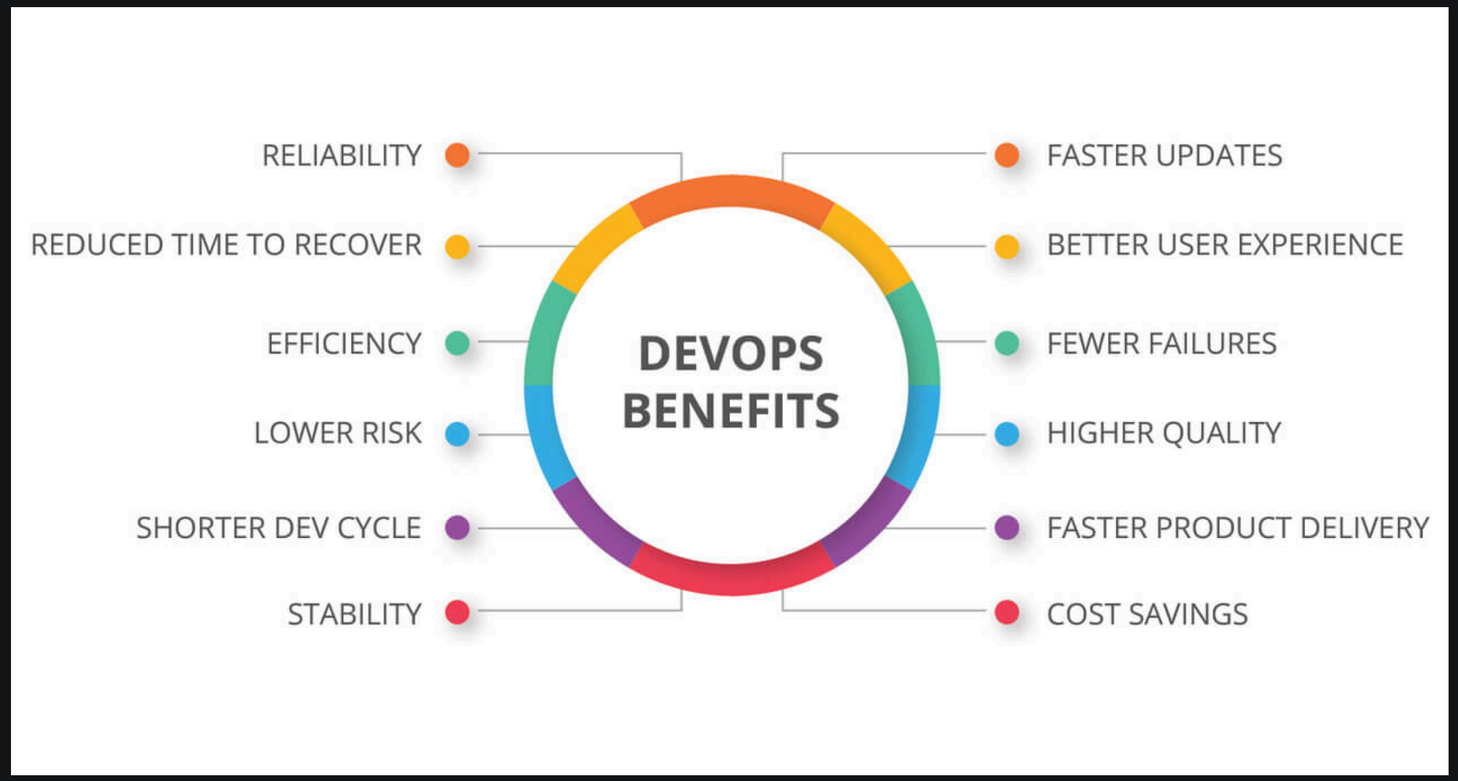
Performance monitoring

Advantages of DevOps Architecture

- Faster deployment
- Continuous delivery
- Automation
- Effective collaboration
- Higher reliability
- Scalability



Disadvantages of DevOps

- New skills required
- Complex architecture
- Security threat
- Infrastructure costs are high

Conclusion :

DevOps Architecture:

- Improves software delivery
- Automates processes
- Encourages collaboration
- Ensures scalability

DevOps architecture is necessary for modern software development

Contribution

Vaani Jindal

- DevOps Architecture Overview & Layers

Harshit Kumar

- DevOps Integration & Pipeline View

Arnav Sharma

- Real-World Example, Advantages & Conclusion

REFERENCES

•AWS DevOps documentation

• Microsoft DevOps guide

• IBM – DevOps Architecture and Practices

INSIGHTS

• “AWS documentation helped us understand the DevOps lifecycle clearly.”

• “Microsoft Azure guide explained CI/CD pipeline in a simple way.”

• “IBM resources helped in understanding advanced DevOps concepts.”

QUESTIONS :

Q1. Define DevOps Architecture?

Ans:- DevOps architecture integrates development and operations to facilitate quicker deployments.

Q2. Mention different DevOps layers?

Ans:- Application layer

Infrastructure layer

CI/CD Layer

Monitoring layer

Security layer

Collaboration layer

Q3. Explain DevOps pipeline.

Ans:- DevOps pipeline is a continuous workflow from building to deployment to monitoring.

Thank You